1). What business and technical skills are critical for today's business intelligence/analytics and information management professionals? As part of InformationWeek's 2014 U.S. IT Salary Survey, business intelligence/analytics and data integration/ warehousing professionals, both staff and managers, were asked to indicate what business and technical skills are critical to their job. The list of business and technical skills included *Analyzing Data*. The following table summarizes the responses to this skill:

	PROFESSIONAL POSITION		
ANALYZING DATA	**Staff**	**Management**	**Total**
Critical	324	255	579
Not critical	86	81	167
Total	410	336	746

Source: Data extracted from "2014 Salary Survey: BI & Analytics," InformationWeek Reports, May 2014, p. 35, **reports.informationweek.com**.

If a professional is selected at random, what is the probability that he or she

a. indicates analyzing data as critical to his or her job?

b. is a manager?

c. indicates analyzing data as critical to his or her job *or* is a manager?

d. Explain the difference in the results in (b) and (c).

2). Do Americans prefer name-brand milk or store brand milk? A survey was conducted by Harris Poll Interactive, Inc. in November 2014. Suppose the results, based on 1,138 females and 1,138 males, were as follows:

	GENDER		
PREFERENCE	**Male**	**Female**	**Total**
Name-Brand Milk	319	341	660
Store Brand Milk	717	649	1,366
Something Else	102	148	250
Total	1,138	1,138	2,276

Source: Data extracted from Harris Interactive, "Do Americans Prefer Name-Brands or Store Brands? Well, That Depends," **bit.ly/1Fpa0KI**.

If an American is selected at random, what is the probability that he or she

a. prefers name-brand milk?

b. is male *and* prefers name-brand milk?

c. is male *or* prefers name-brand milk?

d. Explain the difference in the results in (b) and (c).

3). A survey of 1,700 shoppers asked, "Does social media influence your purchase decisions?" The results indicated that 20.3% of females responded that their purchase decisions are influenced by social media compared to 10.5% of males. (Data extracted from "The Myth of the Mansumer…," **bit.ly/1JVaKoh**.)
The sample sizes of males and females were not provided. Suppose that the results indicated that of 850 males, 89 answered yes. Of 850 females, 173 answered yes. Construct a contingency table to evaluate the probabilities. What is the probability that a respondent chosen at random

a. indicates that social media does influence his/her purchase decisions?

b. is a female *and* indicates that social media does influence his/ her purchase decisions?

c. is a female *or* is a person whose purchase decisions are influenced by social media?

d. is a male *or* a female?

4). How will marketers change their social media use in the near future? A survey by Social Media Examiner of B2B marketers (marketers that focus primarily on attracting businesses) and B2C marketers (marketers that primarily target consumers) was based on 1,184 B2B marketers and 1,703 B2C marketers. The following table summarizes the results:

INCREASE USE OF LINKEDIN?	BUSINESS FOCUS		
	B2B	B2C	Total
Yes	912	920	1,832
No	272	783	1,055
Total	1,184	1,703	2,887

Source: Data extracted from "2014 Social Media Marketing Industry Report," **socialmediaexaminer .com.**

a. Suppose you know that the marketer is a B2B marketer. What is the probability that he or she plans to increase use of LinkedIn?

b. Suppose you know that the marketer is a B2C marketer. What is the probability that he or she plans to increase use of LinkedIn?

c. Are the two events, increase use of LinkedIn and business focus, independent? Explain.

5). Do Americans prefer name-brand milk or store brand milk? A survey was conducted by Harris Poll Interactive, Inc. in November 2014. Suppose the results, based on 1,138 females and 1,138 males, were as follows:

	GENDER		
PREFERENCE	**Male**	**Female**	**Total**
Name-Brand Milk	319	341	660
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Something Else	102	148	250
Total	1,138	1,138	2,276

Source: Data extracted from Harris Interactive, "Do Americans Prefer Name-Brands or Store Brands? Well, That Depends," **bit.ly/1Fpa0KI.**

a. Given that an American is a male, what is the probability that he prefers name-brand milk?

b. Given that an American is a female, what is the probability that she prefers name-brand milk?

c. Is preference independent of gender? Explain.

6). What business and technical skills are critical for today's business intelligence/analytics and information management professionals? As part of InformationWeek's 2014 U.S. IT Salary Survey, business intelligence/analytics and data integration/ warehousing professionals, both staff and managers, were asked to indicate what business and technical skills are critical to their job. The list of business and technical skills included *Analyzing Data*. The following table summarizes the responses to this skill:

	PROFESSIONAL POSITION		
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Source: Data extracted from "2014 Salary Survey: BI & Analytics," InformationWeek Reports, May 2014, p. 35, **reports.informationweek.com**.

a. Given that a professional is staff, what is the probability that the professional indicates analyzing data as critical to his or her job?

b. Given that a professional is staff, what is the probability that the professional does not indicate analyzing data as critical to his or her job?

c. Given that a professional is a manager, what is the probability that the professional indicates analyzing data as critical to his or her job?

d. Given that a professional is a manager, what is the probability that the professional does not indicate analyzing data as critical to his or her job?

Problems

7). A consulting firm submitted a bid for a large research project. The firm's management initially felt they had a 50–50 chance of getting the project. However, the agency to which the bid was submitted subsequently requested additional information on the bid. Past experience indicates that for 75% of the successful bids and 40% of the unsuccessful bids the agency requested additional information.

a. What is the prior probability of the bid being successful (that is, prior to the request for additional information)?

b. What is the conditional probability of a request for additional information given that the bid will ultimately be successful?

c. Compute the posterior probability that the bid will be successful given a request for additional information.

8). A local bank reviewed its credit card policy with the intention of recalling some of its credit cards. In the past approximately 5% of cardholders defaulted, leaving the bank unable to collect the outstanding balance. Hence, management established a prior probability of .05 that any particular cardholder will default. The bank also found that the probability of missing a monthly payment is .20 for customers who do not default. Of course, the probability of missing a monthly payment for those who default is 1.

a. Given that a customer missed one or more monthly payments, compute the posterior probability that the customer will default.

b. The bank would like to recall its card if the probability that a customer will default is greater than .20. Should the bank recall its card if the customer misses a monthly payment? Why or why not?

9). Two Wharton professors analyzed 1,613,234 putts by golfers on the Professional Golfers Association (PGA) Tour and found that 983,764 of the putts were made and 629,470 of the putts were missed. Further analysis showed that for putts that were made, 64.0% of the time the player was attempting to make a par putt and 18.8% of the time the player was attempting to make a birdie putt. And, for putts that were missed, 20.3% of the time the player was attempting to make a par putt and 73.4% of the time the player was attempting to make a birdie putt. (D. G. Pope and M. E. Schweitzer, Is Tiger Woods Loss Averse? Persistent Bias in the Face of Experience, Competition, and High Stakes, June 2009, The Wharton School, University of Pennsylvania).

a. What is the probability that a PGA Tour player makes a putt?

b. Suppose that a PGA Tour player has a putt for par. What is the probability that the player will make the putt?

c. Suppose that a PGA Tour player has a putt for birdie. What is the probability that the player will make the putt?

d. Comment on the differences in the probabilities computed in parts (b) and (c).